Chapter 2. <u>Examples of distribution laws of random variables</u>

http://www.simumath.com/library/book.html?code=Mat_Stat_random_values

2.1 Density function and probability distribution function

Variable is called *random* if as result of experience
it can accept valid values with certain probabilities.

Law of distribution is function (given by table, graph or formula),
allowing to define probability of that random variable

A random variable can be discrete

$$P\{X = x_i\} = p_i, \; i = 1,2,3,\ldots \quad \sum_{i=1}^{\infty} p_i = 1, \qquad (1)$$

The sum of all probabilities is one.

A random variable can be continuous if the probability of falling into the interval (a, b)

$$P\{a \le X \le b\} = \int_a^b f(x)\,dx. \qquad (2)$$

Function *f* (*x*) is called *density function* of continuous random variable.

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$ and $$f(x) \ge 0$$

The sum of all probabilities is one.

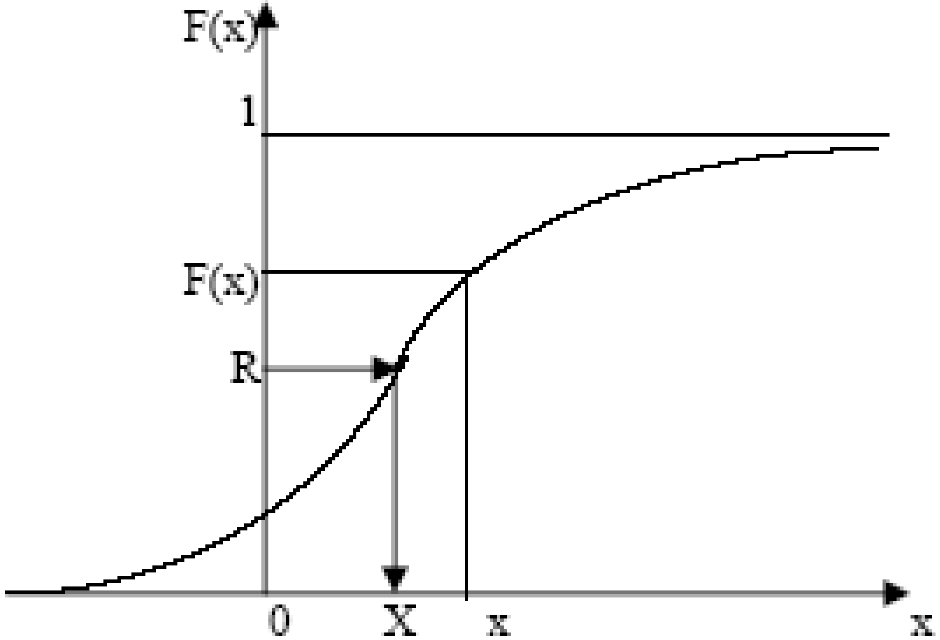
Probability of random variable *X* (discrete or continuous) accepts value, smaller *x*, is called ***distribution function*** of random variable *X* and is designated ***F*(*x*):**

$$F(x) = P\{X < x\}$$ ***distribution function***

For a continuous random variable, the probability distribution function is

$$F(x) = P\{X < x\} = \int_{-\infty}^{x} f(x)\,dx.$$

Typical graph of a probability distribution function is



Basic properties of distribution function:

1) $F(x)$ – not decreasing function, i.e. at $x_2 > x_1$ $\quad F(x_2) \geq F(x_1)$,

2) $F(x)$ – limited function: $0 \leq F(x) \leq 1$,

3) $\lim\limits_{x \to -\infty} F(x) = 0$,

4) $\lim\limits_{x \to +\infty} F(x) = 1$,

not less than zero, but not more than one

2.2 Characteristics of the distribution of a random variable

Mathematical expectation **(mean or expected value** $M(X) = E(X) = \mu_X$

of discrete random variable *X* is the sum:

$$M(X) = E(X) = \mu_X =$$

$$x_1 \cdot p_1 + x_2 \cdot p_2 + x_3 \cdot p_3 + \ldots + x_n \cdot p_n = \sum_{i=1}^{n} x_i p_i \qquad 6(a)$$

Mathematical expectation of continuous random variable *X* is the integral from product of its values *x* and its **density function *f*(*x*):**

$$M(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx. \qquad (6b)$$

Mathematical expectation characterizes *average value* of random variable *X* and is the distribution center

Dispersion. *Dispersion* of random variable *X* is the number:

$$D(X) = M\left\{\left[X - M(X)\right]^2\right\} = M(X^2) - \left[M(X)\right]^2. \qquad (8)$$

Proceeding from definitions of dispersion (8) and mathematical expectation (6a) for discrete random variable and (6b) for continuous random variable we'll receive:

$$D(X) = M(X - M) = \sum_{i=1}^{n} \left(X_i - m\right)^2 \cdot P_i \qquad (9)$$

for discrete random variables.

$$D(X) = M(X - M) = \int_{-\infty}^{+\infty} \left(x - m\right)^2 \cdot f(x) \cdot dx \qquad (9)$$

for continuous random variables

Here $m = M(X)$.

Standard deviation: $\sigma = \sqrt{D(X)}. \qquad (11)$

Standard deviation characterizes the **width** of the probability distribution of a random variable.

Median of random variable *X* is its value *Me* for which the equality

$$P\{X < Me\} = P\{X > Me\}$$

takes place, i.e. equiprobably

In case of symmetric distribution median and mathematical expectation coincide.

2.3 Examples of distribution laws

Binomial distribution. Discrete random variable *X* has *binomial distribution*, if its possible values 0, 1, 2..., *m*, …, *n*, and probabilities corresponding them are equal to

$$P_m = P(X = m) = C_n^m \cdot p^m \cdot q^{n-m}, \quad (21)$$

where **0 < *p* < 1, *q* = 1 – *p* ; *m* = 0, 1, 2, ... , *n* .**

Here $C_n^m = \dfrac{n!}{m!\cdot(n-m)!}$ is the binomial coefficient,

$n! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n$ is a factorial. For instance $4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24$

Mathematical expectation: $E(X) = \mu_X = \dfrac{q}{p}$

Dispersion: $D(X) = \dfrac{q}{p^2}$

Standard deviation: $\sigma = \dfrac{\sqrt{q}}{p}$

Example 2.1

Construct binomial distribution for *n*=2 and *p*=0.5.

According to the equation (21)

m = 0, 1, 2; q=1-0.5=0.5.

$$P_0 = \frac{2!}{0!\cdot(2-0)!}0.5^0 \cdot 0.5^{2-0} = 0.25$$

$$P_1 = \frac{2!}{1!\cdot(2-1)!}0.5^1 \cdot 0.5^{2-1} = 2\cdot 0.25 = 0.5$$

$$P_2 = \frac{2!}{2!\cdot(2-2)!}0.5^2 \cdot 0.5^{2-2} = 0.25$$

In this way

m	0	1	2
P_m	0.25	0.5	0.25

The sum of all probabilities is
0.25 + 0.5 + 0.25 = 1.
Distribution is valid.

Example 2.2

Construct binomial distribution for $n=3$ and p=0.5.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.

Example 2.3

Construct binomial distribution for $n=4$ and p=0.5.

Find mean or expected value $E(X) = \mu_X$

.Find variance (dispersion) and standard deviation.

Example 2.4

With MAPLE construct binomial distribution for *n=10* and
p=0.25; .
p=0.5;
p=0.75.
Construct histogram.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.

Example 2.5

With MAPLE construct binomial distribution for *n=15* and
p=0.25; .
p=0.5;
p=0.75.
Construct histogram.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.

2.3.2 Poisson's distribution.

Discrete random variable X has *Poisson's distribution* if it has infinite calculating set of possible values 0, 1, 2..., m, …, and probabilities corresponding them are defined by the formula:

$$P_m = \frac{\lambda^m \cdot \exp(-\lambda)}{m!}, \qquad m = 0,1,2,\ldots \qquad (23)$$

n = max(m) > 2λ

Examples of the casual phenomena, subordinated to Poisson's distribution law, are: sequence of radioactive disintegration of particles, sequence of refusals at work of complex computer system, a stream of applications on a telephone exchange and many other things.

Poisson's distribution law (23) depends on one parameter λ which simultaneously is both a mathematical expectation, and a dispersion of random variable X . Thus, for Poisson's distribution law the following basic numerical characteristics take place:

Mathematical expectation: $E(X) = \mu_X = \lambda,$

Dispersion: $D(X) = \lambda$ ***Standard deviation:*** $\sigma = \sqrt{\lambda}$

Example 2.6

With MAPLE construct *Poisson's* distribution for *n=15* and

λ=0.5;

λ=5.;

λ=15.;

Construct histogram.

Find mean or expected value $E(X) = \mu_X$

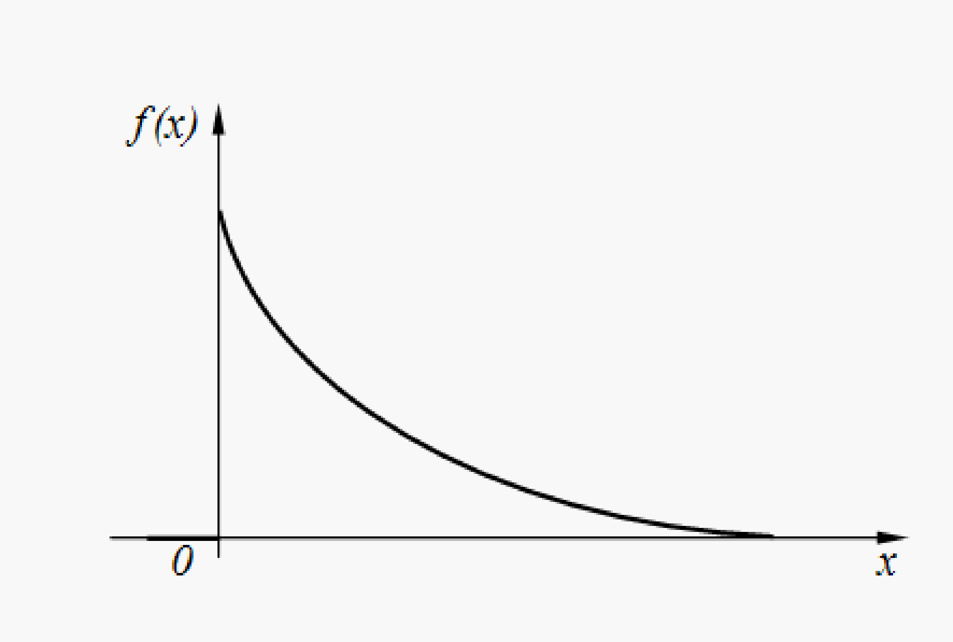
Find variance (dispersion) and standard deviation.

2.3.3 Exponent distribution.

Continuous random variable *X* has an *exponent distribution* if its density function is expressed by the formula:

$$f(x) = \begin{cases} \lambda \cdot e^{-\lambda \cdot x} & at\ x > 0 \\ 0 & at\ x < 0 \end{cases} \qquad (31)$$

Graph of density function (31) is represented on Fig.

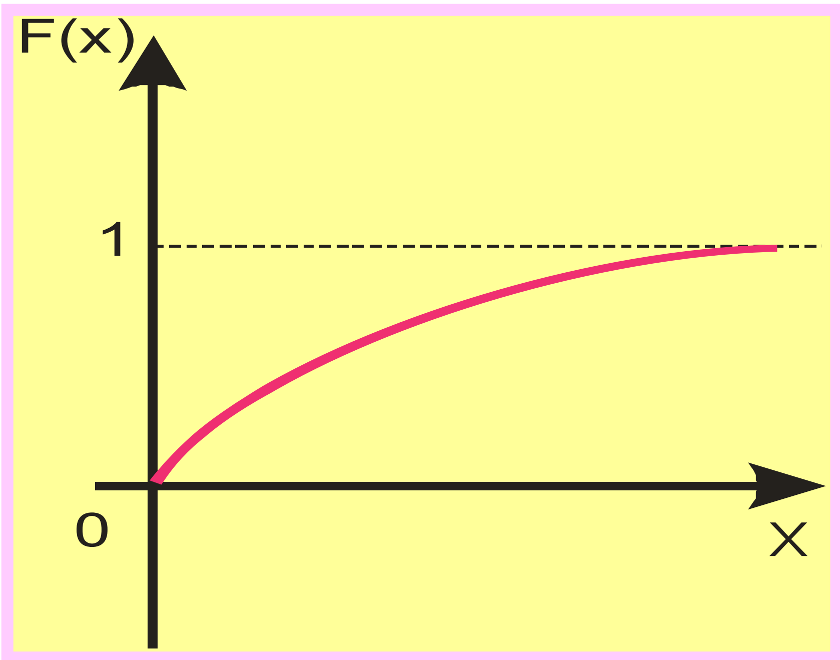


Distribution function of *exponent distribution* law $F(x)$ is:

$$F(x) = P\{X < x\} = \int_{-\infty}^{x} f(x)\,dx$$

$$= \int_{0}^{x} \lambda \cdot e^{-\lambda \cdot x}\,dx = 1 - e^{-\lambda \cdot x}$$

Graph of **d*istribution function*** is represented on Fig.



Time *T* of non-failure operation of computer system is a random variable having exponent distribution with parameter *λ*, which physical sense – an average number of refusals in unit of time.

T is the average operating time of a computer or any other device.

Mathematical expectation: $E(X) = \mu_X = \frac{1}{\lambda} = T,$

Dispersion: $D(X) = \frac{1}{\lambda}$

Standard deviation: $\sigma = \sqrt{\frac{1}{\lambda}}$

$\lambda = \frac{1}{10}$

Example 2.7

Random variable X - the operating time of the refrigerator has an exponential distribution. Find the probability that the refrigerator will work for at least 20 years if the average operating time of the refrigerator is 10 years?

SOLUTION $T = \frac{1}{\lambda} = 10$ $\lambda = \frac{1}{10}$

Using the distribution function, we find the probability that the refrigerator will work for less than 20 years

$$F(x) = P\{X < x\} = 1 - e^{-\lambda \cdot x}$$

$$F(x) = P\{X < 20\} = F(20) = 1 - e^{-\frac{20}{10}}$$

The probability that the refrigerator will work for more than 20 years is found as the probability of the opposite event

$$P(X \geq 20) = 1 - P(X < 20) = 1 - F(20) =$$

$$= 1 - (1 - e^{-\frac{20}{10}}) = e^{-2} \approx 0{,}135$$

Example 2.8

The time for the repair of the device has an exponential distribution law with an average time of 5 hours.
What is the probability of a repair in less than two hours?
What is the probability of repair more than 3 hours?

2.3.4 Normal (Gaussian)distribution

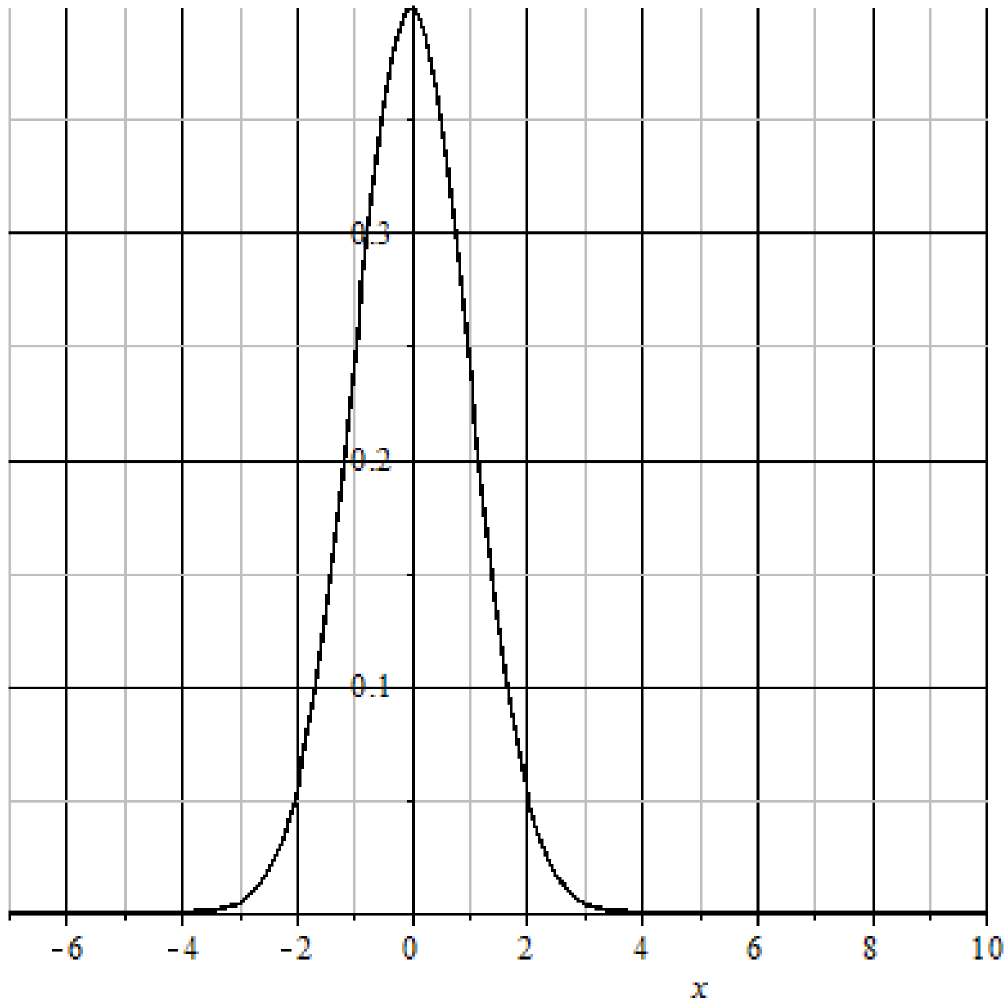
Random variable *X* has *normal (Gaussian) distribution* if density function is defined by the dependence:

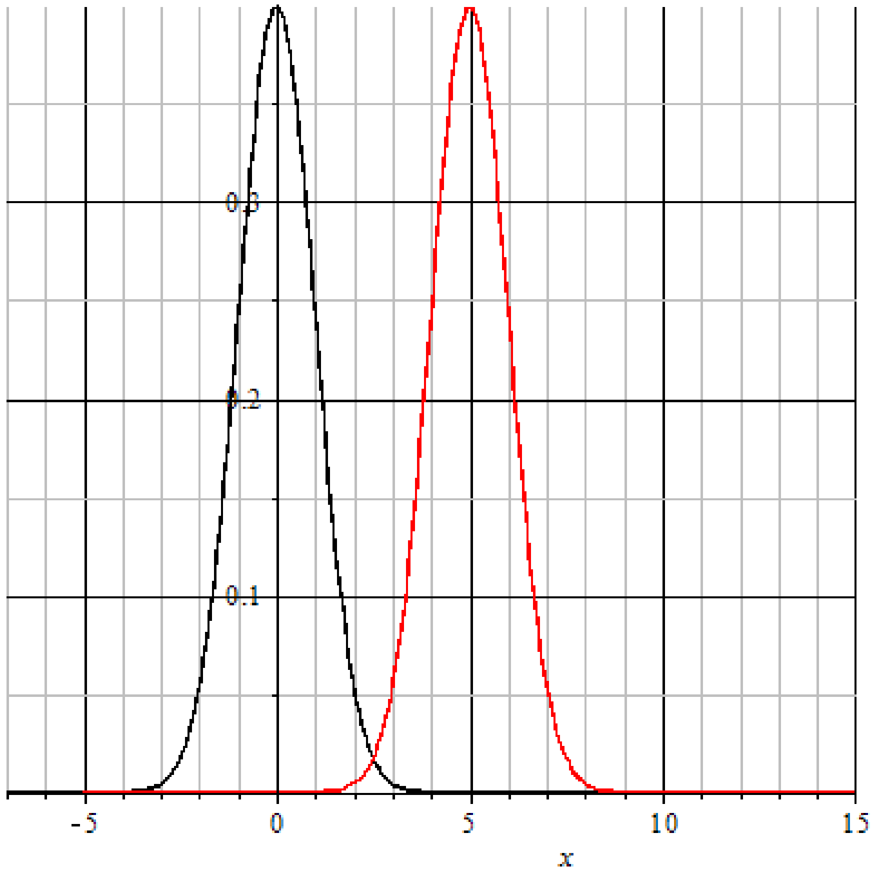
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu_X)^2}{2\sigma^2}}. \qquad (32)$$

where *Mathematical expectation:* $E(X) = \mu_X$,

Standard deviation:σ

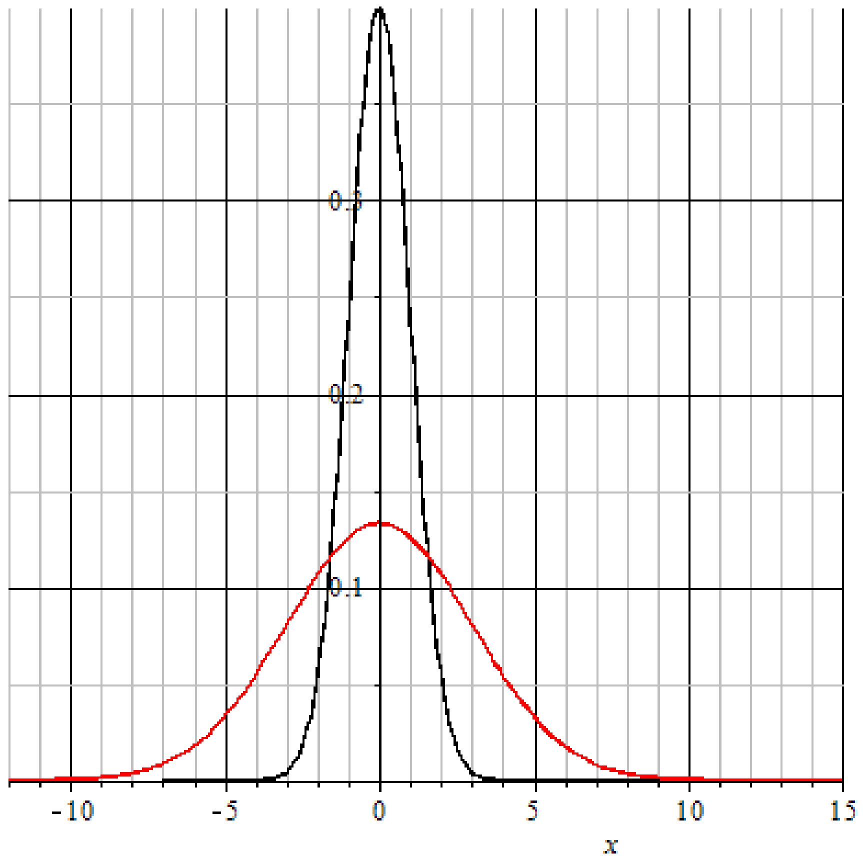
AT $\mu_X = 0, \sigma = 1$ normal distribution is called *standard*. Graph of normal distribution density function (32) is represented on Fig.


0.3
0.2
0.1
-6
-4
-2
0
2
4
6
8
10
x



The black graph corresponds to the expected value µx equal to zero, and the red to the expected value µx equal to five.

The expected value corresponds to the maximum of the density function.



The black graph corresponds to sigma equal to one, and the red sigma equal to 3.

The height and width of the graph depends on sigma: the smaller the sigma, the higher and narrower the graph.

Normal distribution is the most often meeting in various casual natural phenomena. So, mistakes of performance of commands by automated device, mistakes of conclusion of spacecraft in the given point of space, mistakes of parameters of computer systems, etc. in the most cases have normal or close to normal distribution.

Moreover, random variables formed by summing of great quantity of random addendums, are distributed practically according to normal law.

Example 2.9

A continuous random variable has a normal distribution with μx =0, σ = 1.

a) With MIPLE find the probability in the interval [–∞, μx + σ]

b) With MIPLE find the probability in the interval [μx – σ, μx + σ].

c) With MIPLE find the probability in the interval [μx + σ, +∞].

d) Find Pa + Pb + Pc

SOLUTION

$$P(\mu_X - \sigma < X < \mu_X + \sigma) = \int_{\mu_X - \sigma}^{\mu_X + \sigma} f(x)\,dx$$

$$Pb = P(\mu_X - \sigma < X < \mu_X + \sigma) = 0.683$$

This is the one sigma rule.

$$P(-\infty < X < \mu_X - \sigma) = \int_{-\infty}^{\mu_X - \sigma} f(x)\,dx$$

$$Pa = P(-\infty < X < \mu_X - \sigma) = 0.159$$

Example 2.10

A continuous random variable has a normal distribution with $\mu x = 0$, $\sigma = 1$.

a) With MIPLE find the probability in the interval $[-\infty, \ \mu x + 3\sigma]$

b) With MIPLE find the probability in the interval $[\mu x - 3\sigma, \ \mu x + 3\sigma]$.

c) With MIPLE find the probability in the interval $[\mu x + 3\sigma, \ +\infty]$.

d) Find Pa + Pb + Pc